

Age-Structured Transition Analysis of Competitive Exit in U.S. Fencing: Continuation, Temporary Absence, Return, and Terminal Non-Return

Jeremiah Liu

Lexington High School, Lexington, Massachusetts

Keywords: fencing; competitive exit; dropout dynamics; temporary absence; return; adolescent vulnerability; youth sport retention

Abstract

Youth sport dropout is often described by the last age of participation, but competitive participation changes over time. An athlete may continue from season to season, become inactive for a period, return later, or leave competition completely. This study examined competitive exit in U.S. fencing using an age-structured, observed-state transition analysis. We analyzed 13,206 USA Fencing athlete records from 2017 to 2025 and built an athlete-age panel in which each active season was followed by one of three next outcomes: continuation to the next age, absence followed by later return, or terminal non-return within the available follow-up period. We estimated age-specific transition probabilities by weapon and sex, modeled absence onset and terminal non-return across age, quantified return within 2 years after absence onset, described gap duration among returners, and calculated approximate expected remaining active seasons. Continuation was most common at younger ages in all three weapons, but it declined from adolescence onward. Terminal non-return increased from late adolescence into early adulthood. Temporary absences occurred across the age range but were less common than continuation or terminal non-return. Return was more likely when absence began earlier, and most recoverable gaps lasted only 1 year. Expected remaining active seasons decreased with starting age and were generally longest in épée. These results show that competitive exit in fencing is better understood as an age-structured transition process than as a single quitting age.

Introduction

Organized youth sport is widely recognized as an important context for physical, psychological, and social development, yet sustained participation through adolescence remains difficult to maintain (Battaglia et al., 2024; Butcher et al., 2002; Crane & Temple, 2015; Fraser-

Thomas et al., 2008). Across sports, adolescence is repeatedly identified as the period in which participation becomes most vulnerable to withdrawal, as training demands intensify and academic, social, financial, and motivational pressures accumulate (Battaglia et al., 2024; Butcher et al., 2002; Crane & Temple, 2015; Fraser-Thomas et al., 2008). At the same time, recent work has emphasized that dropout is not a single, self-evident event. Its measured rate depends strongly on how withdrawal is defined, what form of participation is being tracked, and whether temporary inactivity is distinguished from permanent exit (Battaglia et al., 2024). These definitional issues are not minor technical details; they shape how dropout is interpreted and how retention strategies are designed.

Fencing provides a particularly informative setting in which to study competitive exit. It is an individual, one-on-one sport organized into three weapons—foil, épée, and saber—that differ in rules, tactical experience, and competitive culture. Long-term progression typically depends on continued club access, coaching, travel, tournament entry, and equipment investment, all of which can amplify dropout pressure during adolescence. Because fencing results are direct and individual, competitive experiences may shape disengagement differently than in many team sports. For some athletes, disengagement may occur as a final one-step exit; for others, it may pass through a period of temporary inactivity before return or before permanent non-return.

Previous research has documented how youth sport dropout typically develops. Studies in organized youth sport consistently identify adolescence as the period of greatest vulnerability, linking withdrawal to intensified specialization, changing self-perceptions of competence, competing obligations, and shifts in motivational climate (Battaglia et al., 2024; Butcher et al., 2002; Crane & Temple, 2015; Fraser-Thomas et al., 2008). Prior research suggests that perceived competence and social context are important for youth sport retention, including in studies of adolescent girls' withdrawal from physical activity (Slater & Tiggemann, 2010; Weiss & Williams, 2004). More generally, review papers have argued that dropout is not only an individual-level motivational problem but also a structural one, shaped by access, culture, opportunity, and system design (Balish et al., 2014; Battaglia et al., 2024). These findings are highly relevant to fencing, which combines individualized outcomes with substantial logistical and financial demands.

However, most dropout analyses remain descriptive, centered on final participation age or last observed season. Such summaries are useful, but they do not reveal whether competitive exit is

driven primarily by a breakdown of year-to-year continuation, a surge of temporary absence, or rising permanent non-return after temporary disengagement. Nor do they show whether age-related vulnerability is driven by direct disappearance from competition or by interruptions that become harder to reverse over time. Those are dynamic questions, and they require a transition-based framework.

The present study applies an observed-state, age-structured analysis to U.S. fencing competition histories. Rather than asking only when athletes are last observed, it asks how the age-structured transition system itself changes across development. Specifically, this paper addresses four questions. First, for an athlete who is competitively active at age a , what is the probability of continuation, temporary absence with later return, or terminal non-return? Second, how do the age patterns of absence onset and terminal non-return vary across weapon and sex? Third, once absence begins, how likely is return within a fixed time horizon, and how long do recoverable gaps tend to last? Fourth, what do the observed transition dynamics imply for expected remaining active seasons across weapon, sex, and starting age?

These questions are mathematically and practically important. A descriptive age distribution of final participation may indicate where exit is concentrated, but it does not reveal whether the underlying issue is a breakdown of continuation, a surge of temporary absence, or rising terminal non-return after temporary disengagement. By modeling athlete histories as an age-structured observed transition system, the present study aims to provide a more mechanistic account of competitive exit in fencing and to clarify how continuation, temporary absence, return, and terminal non-return change across development.

Materials and Methods

Data source and cohort

The study cohort was defined using USA Fencing membership data from the 2020–2021 season. A cohort identified before the end of the observation window was needed so that athletes could be followed long enough to distinguish continuation, temporary absence, and terminal non-return in later competition records. Competition histories for cohort members were then linked across the 2017–2025 seasons to reconstruct age-specific participation records. The final dataset included 13,206 athletes distributed across the three weapons—Foil, Épée, and Saber—and the two gender categories used in the dataset. **Table 1** summarizes the composition of the unique-

athlete cohort, with age defined as 2025–birth year; the transition analyses below are based on athlete-age observations derived from each athlete’s linked competition history in FencingTracker.

Table 1. Composition of the unique-athlete cohort by age in 2025, weapon, and sex (n = 13,206).

Age	Foil M	Foil F	Épée M	Épée F	Saber M	Saber F	Total
11	120	67	49	24	55	30	345
12	207	124	96	65	98	64	654
13	232	182	119	59	148	96	836
14	314	212	147	104	201	120	1,098
15	318	227	163	100	246	126	1,180
16	349	206	192	122	241	131	1,241
17	404	233	232	150	246	150	1,415
18	364	232	232	143	253	167	1,391
19	362	197	237	128	220	119	1,263
20	250	167	183	101	187	92	980
21	176	103	121	75	109	71	655
22	130	72	89	39	83	52	465
23	122	73	82	39	78	52	446
24	81	59	70	28	61	39	338
25	76	26	47	20	35	19	223
26	43	16	30	8	22	10	129
27	63	16	29	10	22	6	146
28	37	21	27	12	17	6	120
29	29	11	28	5	19	4	96
30	28	9	15	5	24	11	92
31	37	9	23	7	10	7	93
Total	3,742	2,262	2,211	1,244	2,375	1,372	13,206

Weapon affiliation

Athletes may appear across more than one weapon over the course of a multi-year career. To ensure consistent subgroup assignment, each athlete was assigned a primary weapon corresponding to the discipline in which they competed most frequently across the observed record. This allowed all athlete-age observations for that athlete to be analyzed within a single weapon group.

Athlete-age panel construction

The data were converted into a long athlete–age panel. A non-missing competition record at a given age was used to classify the athlete as competitively active at that age. Each such record created one active athlete-age observation:

(i, a) : athlete i active at age a .

This yielded an age-by-season participation history for each athlete. These athlete-age records formed the base panel from which the transition outcomes below were constructed.

Observed-state transition definition

For each active athlete-age observation, the next competitive path was classified using only empirically observable outcomes. Let $Y_{i,a}$ denote the observed next-step outcome after athlete i is active at age a . Three categories were defined:

- *Continue* (C): the athlete was active again at age $a + 1$.
- *GapReturn* (G): the athlete was inactive at age $a + 1$ but returned to competition at a later age.
- *TerminalNonReturn* (E): the athlete had no later observed competitive activity within the observable follow-up window.

These categories define the observed transition decomposition:

$$Y_{i,a} \in \{C, E, G\}.$$

Let A_a denote the condition that an athlete is competitively active at age a . For each weapon-sex subgroup, age-specific transition probabilities were estimated as

$$p_C(a) = P(Y = C|A_a), p_G(a) = P(Y = G|A_a), p_E(a) = P(Y = E|A_a),$$

with

$$p_C(a) + p_G(a) + p_E(a) = 1.$$

This is an observed-state system, not a fully identified latent-state absorbing Markov chain. Recoverable gaps are directly observed, but latent temporary disengagement that never returns cannot be separated cleanly from direct terminal exit using the current dataset alone.

Absence onset

Because recoverable gap probability combines both the onset of inactivity and the later possibility of return, absence onset was also modeled directly. Absence onset at age $a + 1$ was defined as

$$\text{AbsenceOnset}_{i,a} = 1$$

if athlete i was active at age a and inactive at age $a + 1$. This outcome includes both athletes who later return and athletes who do not. The age-specific absence-onset probability was estimated as

$$p_A(a) = P(\text{AbsenceOnset} = 1|A_a).$$

Terminal non-return and right-censoring

Terminal non-return was defined conservatively to reduce false labeling of temporary inactivity as permanent exit. Because the observation window ends in 2025, later ages near the right edge of the panel are vulnerable to incomplete follow-up. To address this, terminal non-return was assigned only when sufficient later observation time existed to determine that no later competitive return was present under the selected follow-up rule. If there was not enough future data, those observations were treated as incomplete when analyzing whether athletes had permanently exited the observed competition record.

Smoothed age patterns and vulnerability-window fitting

Age-specific observed probabilities were first computed directly from the athlete-age transition counts. To aid visual interpretation, smoothed age trends were then fitted. For terminal non-return, an age-localized vulnerability-window model was used:

$$B(a, c, w) = \exp\left(-\frac{(a-c)^2}{2w^2}\right),$$

where c is the center of vulnerability and w is its width. When the parametric fit proved unstable or poorly identified for a subgroup, a nonparametric smoother was used instead, and the curve was interpreted descriptively rather than mechanistically. For absence onset, nonparametric smoothing was used to show the age trend, focusing on visualizing the developmental pattern rather than fitting a strict parametric model.

Return within 2 years after absence onset

A recoverable absence episode was defined when an athlete was active at age a , inactive at age $a + 1$, and then returned to competition later. To minimize ambiguity from varying follow-up times, return was analyzed within a fixed time frame:

$$r_2(a) = P(\text{return within 2 years} | \text{absence onset after age } a).$$

Return probabilities were summarized by absence-onset age bins, with 95% confidence intervals.

Gap-duration distribution among returners

For athletes who returned after a competitive absence, gap duration was defined as the number of consecutive inactive years before reappearance in the record. The empirical distribution of recoverable gap duration was then estimated separately by weapon and sex.

Approximate expected remaining active seasons

To summarize the expected future length of competitive participation, we estimated the approximate number of active seasons remaining for a fencer who was active at age a . Let $p_c(a)$ denote the observed probability of direct continuation from age a to age $a + 1$, conditional on being active at age a . Let $p_{G,d}(a)$ denote the observed probability, again conditional on being active at age a , of returning after a temporary absence of d inactive years. Terminal non-return contributes no future active seasons beyond the current one and therefore enters the recursion only through the remaining probability mass. The approximate expected remaining active seasons, $E(a)$, were then calculated recursively as

$$E(a) = 1 + p_c(a) E(a + 1) + \sum_{d \geq 1} p_{G,d}(a) E(a + d + 1),$$

where the leading 1 represents the current active season. The summation is taken over the gap durations supported in the observed data. This recursion was evaluated backward from older ages to younger ages within each weapon-sex subgroup using the observed continuation and return structure.

Statistical visualization

All analyses were carried out using MATLAB. Observed probabilities were shown with 95% confidence intervals where applicable. For return probability, age-binned estimates were used to stabilize inference in sparse older-age ranges.

Results

Observed transition decomposition from active competition

The observed transition decomposition revealed a clear age-structured competitive pathway in all three weapons (**Figure 1**). At younger ages, continuation overwhelmingly dominated the next-step dynamics. In every subgroup, the probability of Continue remained high through childhood and early adolescence, indicating that once athletes were competitively active, most remained active at the next age.

That continuation dominance weakened progressively with age. In foil and saber, continuation began to erode in the mid-teen years and declined further into the twenties. Épée showed a more persistent continuation profile, with high continuation extending farther into adolescence before declining more gradually. This weapon contrast is consistent with the broader impression that épée may support a somewhat more durable competitive pathway than foil or saber.

The complementary rise occurred primarily in *TerminalNonReturn*, not in *GapReturn*. Across all subgroups, terminal non-return remained low at younger ages, then increased markedly from adolescence onward. The age at which this increase became pronounced varied somewhat by weapon and gender, but the overall pattern was consistent: the main observable change in the transition system was a shift away from continuation toward terminal disappearance from the competitive record. Recoverable gaps occurred across the age range, but they were much less common than either continuation or terminal non-return.

This distinction is important. A final-age distribution alone cannot tell whether athletes are increasingly disappearing through recoverable inactivity or direct terminal non-return. The transition decomposition shows that competitive exit in fencing is dominated by the latter. Temporary absence is present, but it is not the main pathway driving age-related dropout in the observed record.

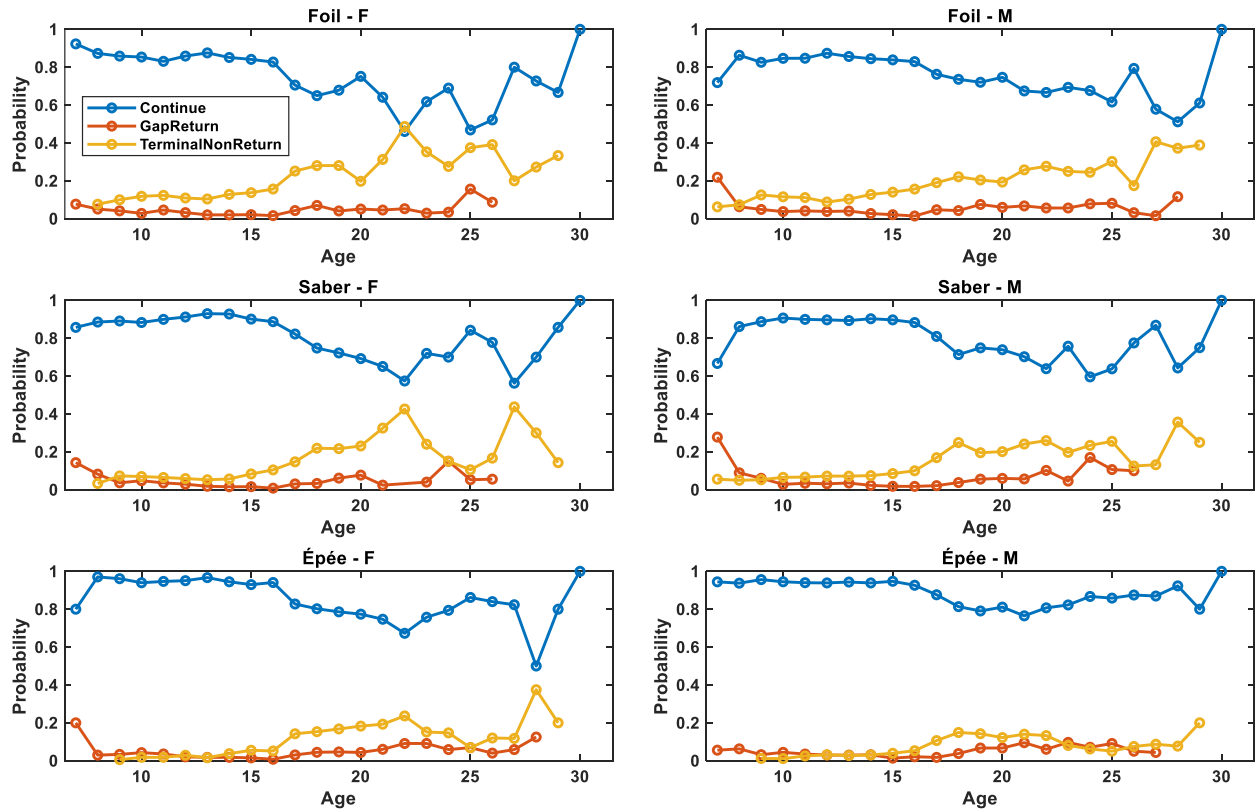


Figure 1: Observed transition decomposition from active competition for (Left column) female and (Right column) male fencers in Foil (top row), Saber (middle row), and Épée (bottom row). Blue circles denote the probability of *Continue*, orange circles denote *GapReturn*, and yellow circles denote *TerminalNonReturn* after an active competitive season at each age.

Absence onset as an age-dependent process

The age-specific probability of absence onset rose steadily from childhood into adolescence and early adulthood in all six subgroups (**Figure 2**). In younger ages, the observed absence-onset probability was low, indicating that next-year inactivity after an active season was relatively uncommon. As age increased, however, absence onset became substantially more likely.

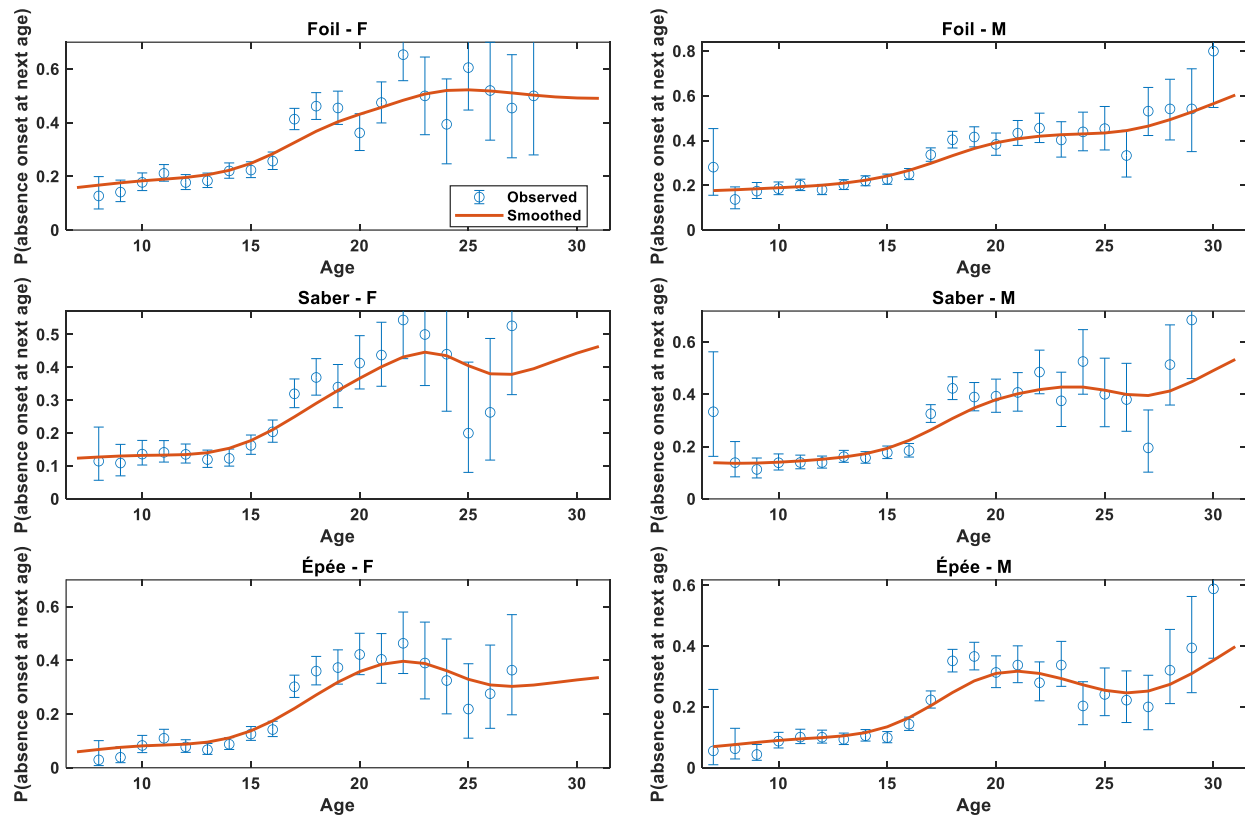


Figure 2: Age-specific probability of absence onset for (Left column) female and (Right column) male fencers in Foil (top row), Saber (middle row), and Épée (bottom row). Open circles show the observed age-specific probabilities, error bars show 95% confidence intervals, and the red curve shows a nonparametric smoothed age trend.

The overall shape was broadly similar across weapon-sex groups, but the levels differed. Foil and saber generally exhibited higher absence-onset probabilities than épée, especially in later adolescence and the early twenties. The smoothed curves showed that the age trend was not purely linear. In several groups, absence onset increased sharply during the later teen years, leveled somewhat in the early twenties, and remained elevated thereafter. The widening uncertainty bars at older ages reflect the smaller number of athletes remaining at risk rather than a breakdown of the overall trend.

These results indicate that the erosion of continuous participation begins before terminal non-return becomes dominant. In other words, athletes initially become more likely to miss a year of competition, and permanent dropout rises only later. This makes absence onset a useful early signal of competitive vulnerability.

Terminal non-return and age-localized vulnerability

Terminal non-return displayed a strong age structure in every subgroup (**Figure 3**). At younger ages, the observed probability of terminal non-return after an active season was low. It then rose into a late-adolescent or early-adult window before flattening or declining at the oldest ages with adequate support.

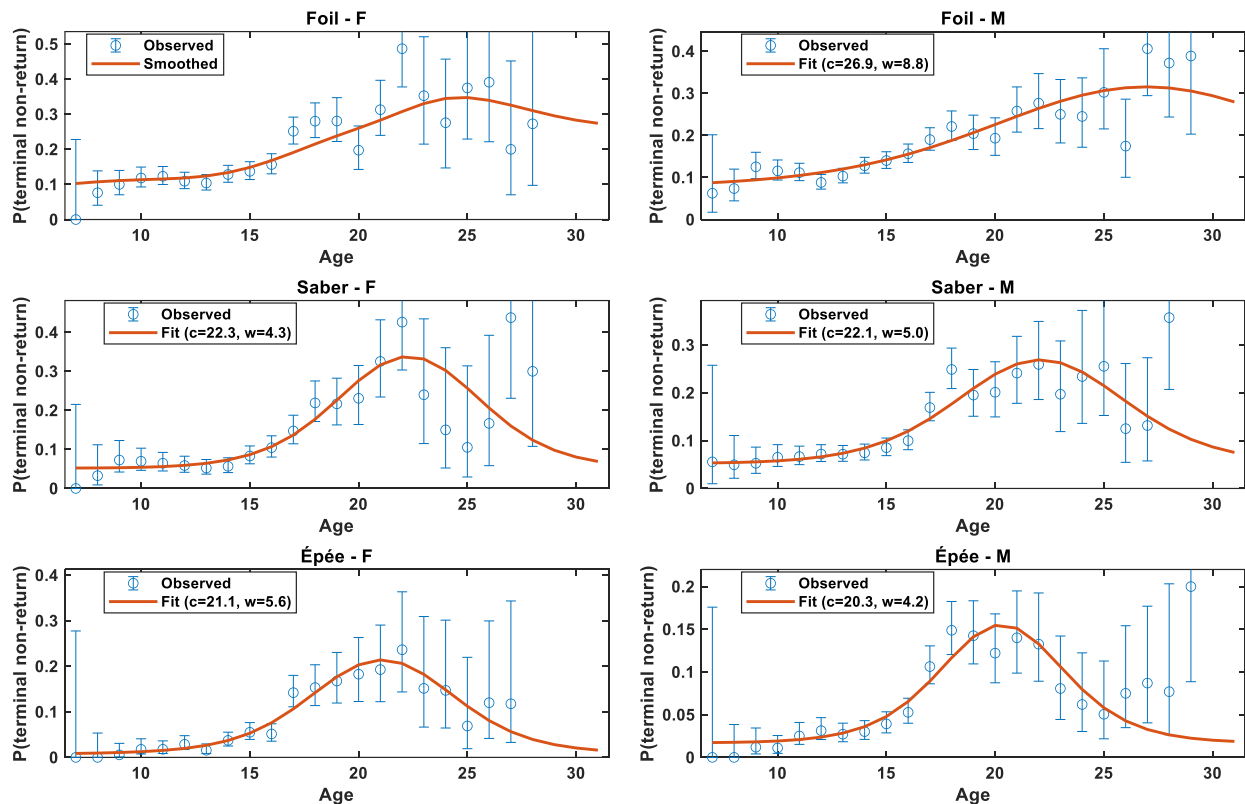


Figure 3: Age-specific probability of terminal non-return for (Left column) female and (Right column) male fencers in Foil (top row), Saber (middle row), and Épée (bottom row). Open circles show the observed age-specific terminal non-return probabilities, error bars show 95% confidence intervals, and the red curve shows either a fitted vulnerability-window model or a nonparametric smoother when the parametric fit was unstable.

In five of the six subgroups, the age pattern was sufficiently stable to support an interpretable vulnerability-window fit. The estimated centers in these stable subgroups clustered in the late-adolescent to early-adult range, roughly from the low twenties to the high twenties, with moderate widths. Épée males displayed the lowest overall terminal non-return levels and the narrowest fitted

window among the stable parametric fits, whereas foil and saber generally showed broader or higher-risk windows. For foil females, the parametric fit was unstable, and a nonparametric smoother was used instead. That instability itself suggests that the age pattern in that subgroup is better described as a broad developmental rise than as a sharply localized parametric window. The terminal non-return curves therefore identify the clearest age-localized vulnerability feature in the dataset. Adolescence does not merely coincide with more athletes being last observed; rather, it marks the developmental period in which active competitors become much more likely to disappear from the record without later return.

Return within 2 years after absence onset

Temporary disengagement remained a meaningful pathway, even though continuation and terminal non-return were more common overall. **Figure 4** shows the probability of returning within 2 years after absence onset by absence-onset age bin. In general, return was more likely when absence began at younger ages and less likely when it began in mid-adolescence or early adulthood.

The pattern varied somewhat by subgroup. Among foil males, return probability declined gradually across age bins. Foil females showed a less regular pattern, but the overall trend was still downward, with greater uncertainty in the oldest supported bins. Both saber males and saber females had lower return probabilities in the mid- to late-adolescent bins than at younger onset ages. Épée had the highest return probabilities overall, especially among males, consistent with its stronger overall participation continuity.

Using age bins helped stabilize the estimates and reduced overinterpretation of sparse older-age data. Even so, the confidence intervals widened at later ages because fewer athletes initiated observable gaps there. Overall, the result is clear: temporary absence was more likely to be reversible when it began earlier.

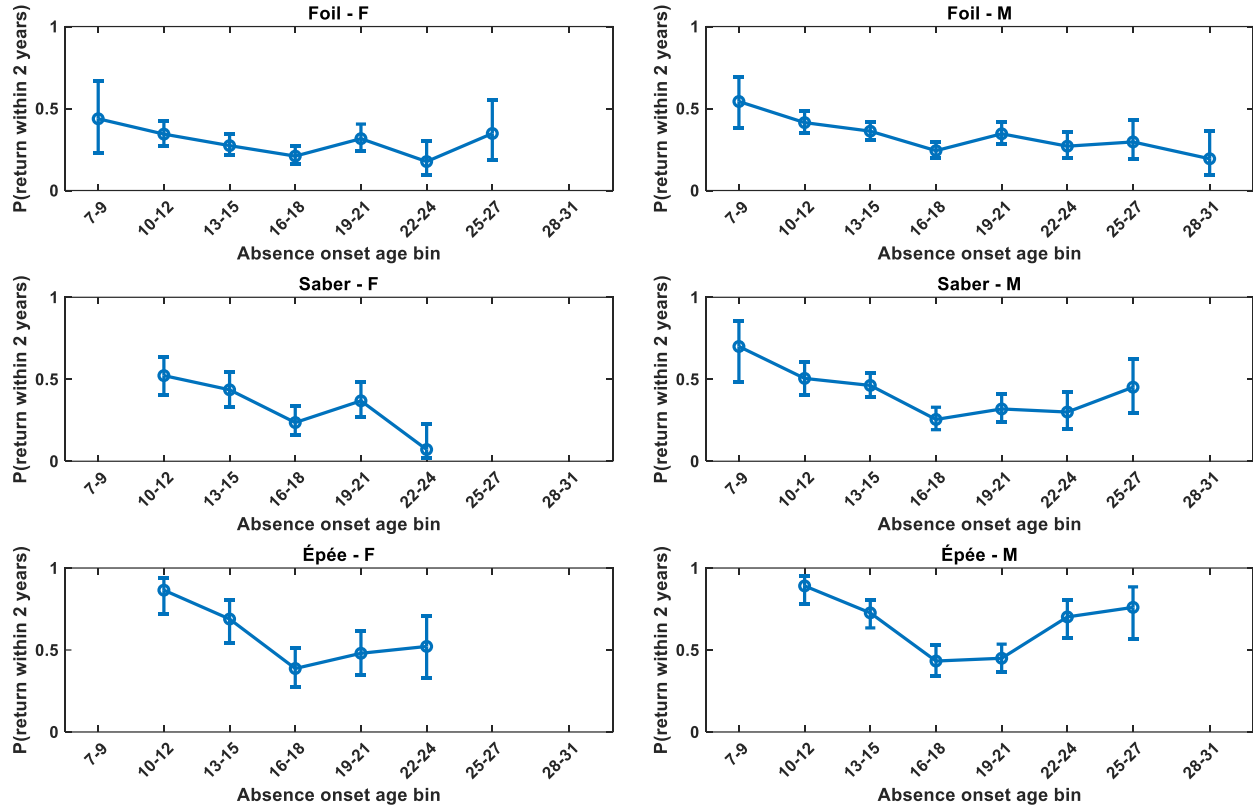


Figure 4: Probability of return within 2 years after absence onset for (Left column) female and (Right column) male fencers in Foil (top row), Saber (middle row), and Épée (bottom row). Open blue circles denote observed bin-level return probabilities by absence-onset age bin, with 95% confidence intervals.

Gap duration among returners

Among athletes who did return after an absence, the overwhelming majority returned after 1-year gaps (**Figure 5**). Two-year gaps were much less common, and longer recoverable absences were rare in every subgroup. This pattern was strikingly consistent across weapon and sex. Foil and épée showed the highest concentration of 1-year recoverable gaps, while saber exhibited a somewhat broader but still strongly 1-year-dominated gap distribution. The rarity of longer recoverable gaps means that competitive return, when it occurs, usually happens quickly. This supports the choice of a 2-year return horizon in the main return analysis: most recoverable behavior is already captured within that time frame.

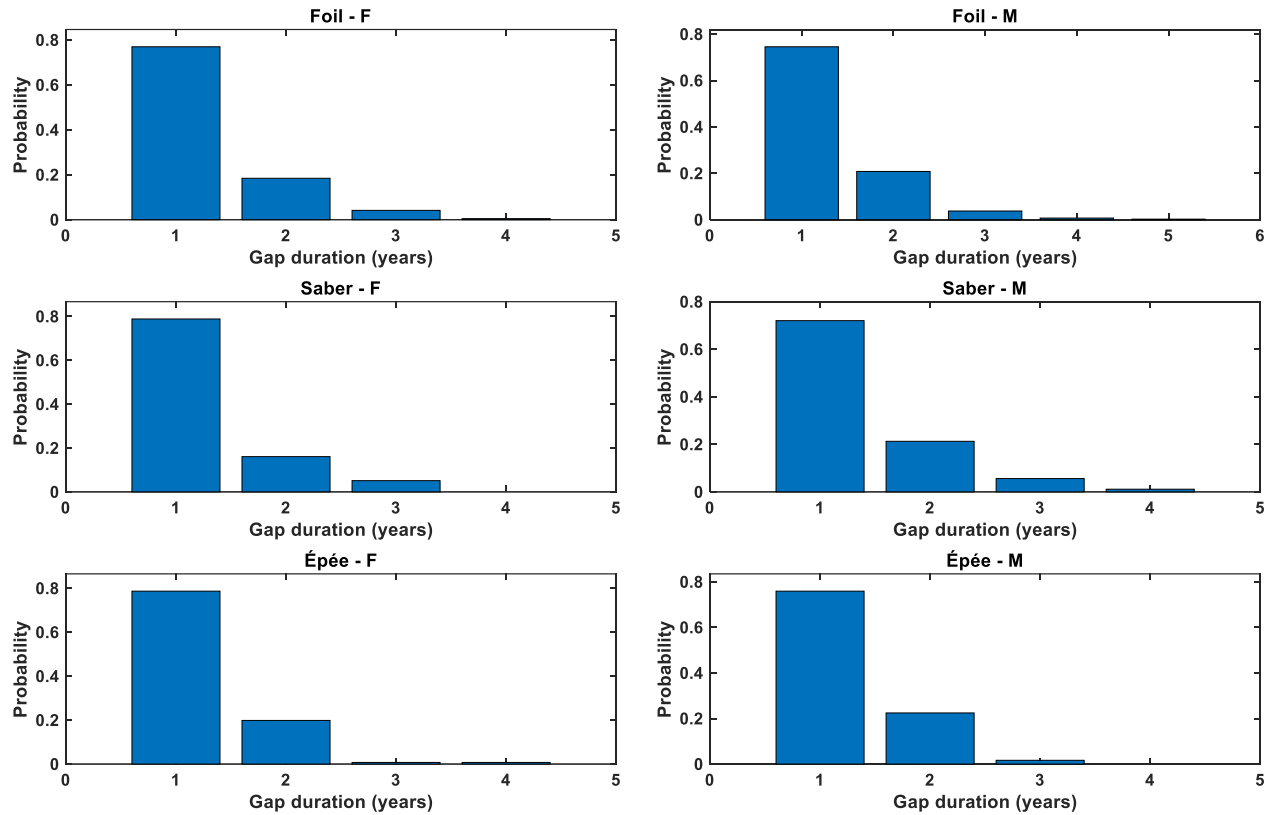


Figure 5: Distribution of gap duration among returners for (Left column) female and (Right column) male fencers in Foil (top row), Saber (middle row), and Épée (bottom row). Blue bars denote the probability that a recoverable competitive absence lasted 1, 2, 3, or more years before return.

Approximate expected remaining active seasons

Using the observed age-specific continuation and return structure, we estimated the approximate number of active competitive seasons remaining for a fencer who was active at a given starting age. This quantity includes the current active season plus future active seasons reached either through direct continuation to the next age or through later return after a temporary absence. The approximate expected number of future active competitive seasons declined monotonically with starting age in all six subgroups (**Figure 6**). This monotonicity is what one would expect in an age-structured competitive system, but the subgroup differences are informative. Épée athletes had the longest expected remaining active seasons at nearly every starting age, followed by saber and then foil. Within each weapon, males generally had slightly higher expected remaining active seasons than females, though the magnitude of the difference varied.

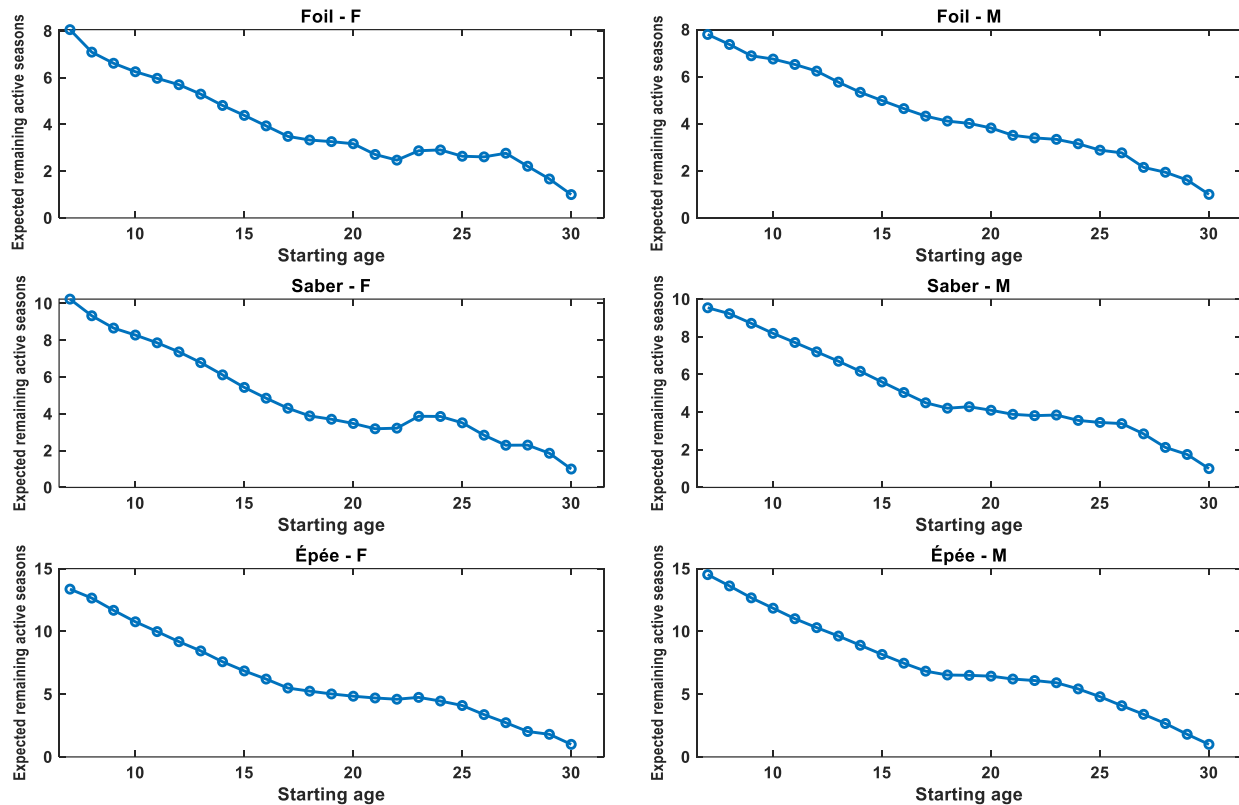


Figure 6: Approximate expected remaining active seasons as a function of starting age for (Left column) female and (Right column) male fencers in Foil (top row), Saber (middle row), and Épée (bottom row). Blue circles denote the estimated number of future active competitive seasons derived from the observed continuation and return structure.

The shape of the decline was not perfectly linear. In several groups, especially saber and épée, the curve flattened somewhat in the late teens or early twenties before continuing downward. This reflects the fact that competitive life expectancy is jointly influenced by continuation, absence onset, and return, not by age alone. Nevertheless, the broad interpretation is straightforward: earlier ages provide more remaining competitive opportunity, and that expected horizon is longest in the subgroup structure with the most persistent continuation and lowest terminal non-return.

Discussion

This study reframed competitive exit in U.S. fencing as an age-structured observed transition system rather than a single quitting age. That change in perspective yielded several important findings.

First, the main structural change with age is the weakening of continuation together with the rise of terminal non-return. In childhood and early adolescence, active fencers in all three weapons usually remained active at the next age. As age increased, that continuity weakened. The transition

decomposition showed that this weakening was not mainly absorbed by a large increase in recoverable gaps. Instead, a growing share of athletes disappeared from the competitive record without later return. This distinction matters because a final-age summary alone cannot show whether athletes are mainly pausing or leaving the competitive pathway altogether. The present results indicate that, in the observed record, the larger shift is toward terminal non-return.

Second, absence onset and terminal non-return followed related but distinct age patterns. Absence onset began rising earlier, from adolescence into early adulthood, indicating that the erosion of year-to-year continuity starts before permanent non-return reaches its highest levels. In this sense, absence onset acts as an earlier sign of competitive vulnerability, whereas terminal non-return is the later and more definitive expression of that vulnerability. The terminal non-return curves further suggest that risk is concentrated in a broad late-adolescent to early-adult window rather than at a single sharply defined quitting age.

Third, temporary absence remained meaningful even though it was not the dominant pathway. Recoverable absences were observed across the age range, and return within 2 years was more likely when absence began earlier. At the same time, most recoverable gaps lasted only 1 year, with 2-year gaps already much less common and longer recoverable gaps rare. This pattern suggests that reversible disengagement in fencing is usually short. When athletes do return, they tend to do so quickly. From a retention standpoint, that implies that early re-engagement efforts may have value, but that the practical window for successful return is limited.

Fourth, weapon differences were consistent across the analyses. *Épée* showed lower absence onset, lower terminal non-return, higher return probabilities, and longer expected remaining active seasons than foil or saber. These differences do not by themselves identify a causal mechanism, but they do suggest that competitive exit is not uniform across fencing weapons. Possible explanations include differences in competitive format, athlete selection, club structure, travel patterns, or developmental culture. Whatever the cause, the results indicate that fencing should not be treated as a single retention environment.

The approximate expected remaining active seasons help connect the age-specific transitions to a more intuitive summary of competitive persistence. These curves declined with starting age in every subgroup, as expected, but the subgroup contrasts were informative. *Épée* athletes generally had the longest remaining competitive horizon, followed by saber and then foil. The curves also showed that future competitive persistence depends on more than age alone: it is

shaped jointly by continuation, absence onset, and the possibility of return after temporary absence.

These findings have several practical implications. Retention work in fencing should focus not only on who eventually leaves, but also on who is beginning to lose year-to-year continuity. The results suggest that adolescence and early adulthood are the key ages at which the competitive pathway becomes less stable. Support during this stage may need to emphasize continuity of participation, rapid response to emerging absences, and low-barrier routes back into competition after short interruptions. Because most recoverable gaps are short, re-entry efforts are likely to be most effective when they occur quickly rather than after long inactivity.

Several limitations should be noted. The analysis is restricted to sanctioned competition histories and therefore does not capture recreational fencing, noncompetitive club participation, or informal involvement in the sport. Terminal non-return from the record is not identical to complete withdrawal from fencing. In addition, the observed-state framework does not fully identify latent inactive states, so the expected remaining active seasons are approximate rather than exact absorbing-chain quantities. Finally, inference becomes less stable at older ages because fewer athletes remain under observation there.

Even with these limitations, the observed-state framework adds something important that descriptive final-age summaries cannot provide. It shows how competitive exit is generated: through declining continuation, rising absence onset, increasing terminal non-return, and only limited recovery once continuity is lost. For retention research in fencing, that age-structured dynamic perspective is likely to be more informative than final participation age alone.

Conclusion

Competitive exit in U.S. fencing is better understood as an age-structured transition process than as a single quitting age. Using 13,206 athlete records, this study showed that continuation dominates at younger ages but weakens from adolescence onward, while terminal non-return rises into late adolescence and early adulthood. Absence onset also increases with age, providing an earlier sign of competitive vulnerability, but most recoverable gaps are short and return is more likely when absence begins earlier. Across nearly all analyses, épée showed the most persistent competitive pathway. These results move fencing dropout research beyond descriptive final-age patterns toward a more mechanistic view of competitive retention. By distinguishing continuation, temporary absence, return, and terminal non-return, the study clarifies how athletes move away

from the competitive pathway and when those changes become most pronounced. This framework provides a stronger basis for understanding competitive exit in fencing and for designing age-targeted strategies to support longer-term participation.

Acknowledgments

The author gratefully acknowledges the availability of USA Fencing competition-history records accessed through FencingTracker, which made this analysis possible.

References

- Balish, S. M., McLaren, C., Rainham, D., & Blanchard, C. (2014). Correlates of youth sport attrition: A review and future directions. *Psychology of Sport & Exercise*, 15(4), 429-439. <https://doi.org/10.1016/j.psychsport.2014.04.003>
- Battaglia, A., Kerr, G., & Tamminen, K. (2024). The Dropout From Youth Sport Crisis: Not as Simple as It Appears. *Kinesiology Review*, 13(3), 345-356. <https://doi.org/10.1123/kr.2023-0024>
- Butcher, J., Lindner, K. J., & Johns, D. P. (2002). Withdrawal from competitive youth sport: a retrospective ten-year study. *Journal of Sport Behavior*, 25, 145+. https://link.gale.com/apps/doc/A86049189/AONE?u=mlin_n_umass&sid=ebsco∓xid=cbd7e3f3
- Crane, J., & Temple, V. (2015). A systematic review of dropout from organized sport among children and youth. *European Physical Education Review*, 21(1), 114-131. <https://doi.org/10.1177/1356336x14555294>
- Fraser-Thomas, J., Côté, J., & Deakin, J. (2008). Understanding dropout and prolonged engagement in adolescent competitive sport. *Psychology of Sport & Exercise*, 9(5), 645-662. <https://doi.org/10.1016/j.psychsport.2007.08.003>
- Slater, A., & Tiggemann, M. (2010). “Uncool to do sport”: A focus group study of adolescent girls’ reasons for withdrawing from physical activity. *Psychology of Sport & Exercise*, 11(6), 619-626. <https://doi.org/10.1016/j.psychsport.2010.07.006>
- Weiss, M. R., & Williams, L. (2004). The Why of Youth Sport Involvement: A Developmental Perspective on Motivational Processes. In *Developmental sport and exercise psychology: A lifespan perspective*. (pp. 223-268). Fitness Information Technology. <https://research.ebsco.com/linkprocessor/plink?id=c01fbbff-f100-3efa-91da-0996e6f6b7f1>